11 - Monthly Project Reviews

Chronological progression from Review 1 to Review 4 (read from the .pptx files)

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

review/review_1.pptx ... review_4.pptx; review/README.md; review/scripts/*.py

The decks were generated by the scripts in review/scripts. Reviews 3 and 4 read real pipeline output; Reviews 1 and 2 describe design and foundations and claim no results. Dates, student name and guide are placeholders in the decks.

Progression at a glance

Review	Theme	Slides	Results shown?
Review 1	Introduction, literature survey, proposed solution	11	no
Review 2	Methodology, architecture, data, preprocessing, initial implementation	9	no
Review 3	Machine learning, anomaly detection, trust scoring, dashboards	10	yes (from pipeline output)
Review 4	Results, comparison, dashboards, conclusions	15	yes (from pipeline output)

Project progression across the four reviews.

Review 1 - Introduction, literature survey, proposed solution

Slide 1: FinTech Cybersecurity and Consumer Trust

- Monthly Project Review 1 - Introduction, Literature Survey and Proposed Solution
- Final-year research & development project
- [Student name] - [Register no.] - Guide: [Guide name] - [Department / Institution]
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Slide 2: Agenda

- FinTech Cybersecurity and Consumer Trust | Review 1 | 1
- - Introduction and motivation
- - Problem statement
- - Aim and objectives
- - Literature survey
- - Research gap
- - Proposed solution and plan
- - Ethics and scope

Slide 3: Introduction and motivation

- FinTech Cybersecurity and Consumer Trust | Review 1 | 2
- - FinTech moves payments, identity and personal data through software that is attacked continuously
- - Providers invest in fraud detection and threat monitoring
- - Adoption depends on consumer trust, which reflects perceived as well as actual security
- - Trust is multi-dimensional: competence, integrity, benevolence (McKnight et al., 2002)
- - Security teams and trust researchers rarely share one reproducible evidence base

Slide 4: Problem statement

- FinTech Cybersecurity and Consumer Trust | Review 1 | 3
- - Fraud-detection studies optimise predictive metrics, often on a single dataset and with accuracy-style reporting
- - Consumer-trust studies rely on surveys that are disconnected from measured security performance
- - Detection errors have a consumer cost: false positives block legitimate customers
- - Research pipelines are seldom observable: results are not monitored or easily reproduced
- Core question: how do measurable cybersecurity capabilities relate to consumer trust, and can the experiment itself be monitored?

Slide 5: Aim and objectives

- FinTech Cybersecurity and Consumer Trust | Review 1 | 4
- ID | Objective
- O1 | Reproducible pipeline for transaction and security-event data (synthetic + optional public)
- O2 | Compare supervised and unsupervised fraud/anomaly detectors with imbalance-aware metrics
- O3 | Quantify the operational cost of detection errors (false-positive rate)
- O4 | Measure consumer trust with a documented, privacy-conscious survey
- O5 | Propose and stress-test a composite Consumer Trust Index
- O6 | Instrument the pipeline so Grafana shows real collected data

Slide 6: Literature survey

- FinTech Cybersecurity and Consumer Trust | Review 1 | 5
- Theme | Key works | Take-away for this project
- ML fraud detection | Ngai et al. 2011; Bhattacharyya et al. 2011; Chen & Guestrin 2016 | Ensembles/boosting are strong baselines
- Feature engineering | Bahnsen et al. 2016 | History/behavioural aggregates matter
- Imbalance and metrics | Davis & Goadrich 2006; Saito & Rehmsmeier 2015; Dal Pozzolo et al. 2015 | Use PR-AUC; avoid resampling that distorts probabilities
- Anomaly detection | Liu et al. 2008; Breunig et al. 2000; Chandola et al. 2009 | Label-free detectors for delayed labels
- Trust in e-commerce | McKnight 2002; Gefen 2003; Pavlou 2003; Davis 1989 | Multi-dimensional trust constructs
- Measurement/privacy | Cronbach 1951; Likert 1932; Sweeney 2002 | Reliability checks; k-anonymity
- References: report/references.bib (entries to be verified against publisher records).

Slide 7: Research gap

- FinTech Cybersecurity and Consumer Trust | Review 1 | 6
- What exists
- - Strong fraud models evaluated in isolation
- - Trust surveys without security telemetry
- - Monitoring treated as an operations topic, not a research instrument
- What is missing
- - Joint view of measured security and perceived trust
- - Transparent, sensitivity-tested composite index
- - Reproducible, observable research platform suitable for replication

Slide 8: Proposed solution - research workflow

- FinTech Cybersecurity and Consumer Trust | Review 1 | 7
- Data (synthetic / public)
- Validate + clean
- Features
- Train + tune models
- Evaluate + risk tiers
- Survey + trust score
- Trust index
- Metrics, logs, dashboards
- Python pipeline + PostgreSQL + JupyterLab, monitored with Prometheus, Loki and Grafana; all in Docker Compose.

Slide 9: Proposed solution - planned architecture

- FinTech Cybersecurity and Consumer Trust | Review 1 | 8
- Docker network: fintech-net (ports bound to localhost)
- Worker
- Python pipeline
- + /metrics :8000
- JupyterLab
- :8888
- PostgreSQL
- results + metadata

- Promtail ->
- Loki (logs)
- Prometheus
- :9090
- cAdvisor
- container stats
- Grafana :3000
- 7 dashboards
- alerts
- Worker writes results to PostgreSQL and JSON logs (-> Promtail -> Loki); Prometheus scrapes worker, cAdvisor, Loki and Grafana; Grafana queries Prometheus (PromQL), Loki (LogQL) and PostgreSQL (SQL). Jupyter reads the artefacts the worker produced.

Slide 10: Plan and expected deliverables

- FinTech Cybersecurity and Consumer Trust | Review 1 | 9
- Month | Focus | Deliverable
- 1 | Introduction, literature, gap, design | This review; repository skeleton
- 2 | Methodology, architecture, data, preprocessing | Generators, validation, cleaning, features, containers
- 3 | Models, anomaly detection, trust scoring, dashboards | Training pipeline, metrics exporter, Grafana
- 4 | Results, comparison, conclusions | Evaluation, report chapters, final demo
- Status at Review 1: planning and design only; no experimental results are claimed in this review.

Slide 11: Ethics, scope and risks

- FinTech Cybersecurity and Consumer Trust | Review 1 | 10
- - Security experiments use synthetic data in an isolated environment, or licensed public datasets; no production systems are probed
- - Survey: voluntary, no direct identifiers, banded demographics, k-anonymity; ethics approval required before real data collection
- - Risk: synthetic data encodes our assumptions - results validate the method, not real prevalence
- - Risk: the trust index is a proposal, not a validated instrument - sensitivity analysis planned

Review 2 - Methodology, architecture, data, preprocessing, initial implementation

Slide 1: FinTech Cybersecurity and Consumer Trust

- Monthly Project Review 2 - Methodology, Architecture, Data and Initial Implementation
- Final-year research & development project
- [Student name] - [Register no.] - Guide: [Guide name] - [Department / Institution]
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Slide 2: Progress since Review 1

- FinTech Cybersecurity and Consumer Trust | Review 2 | 1
- - Objectives and research questions fixed (RQ1-RQ7, hypotheses H1-H7)
- - Repository structure, configuration (settings.yaml) and Docker Compose skeleton created
- - Synthetic data generators, validation, cleaning, feature engineering implemented and unit-tested
- - PostgreSQL schema and Prometheus exporter drafted
- - Not yet in scope for this review: model comparison, dashboards, trust results

Slide 3: Research methodology workflow

- FinTech Cybersecurity and Consumer Trust | Review 2 | 2
- Seeded data generation
- Validation (13 checks)
- Cleaning
- Leakage-safe features
- Chronological split 70/15/15
- Tune on train, threshold on validation
- Single test evaluation
- Statistics + trust analysis
- Chronological splitting prevents the model from learning from the future; the test set is used once.

Slide 4: System architecture (Docker Compose)

- FinTech Cybersecurity and Consumer Trust | Review 2 | 3
- Docker network: fintech-net (ports bound to localhost)
- Worker
- Python pipeline
- + /metrics :8000
- JupyterLab
- :8888
- PostgreSQL
- results + metadata
- Promtail ->
- Loki (logs)
- Prometheus
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Slide 5: Technology stack

- FinTech Cybersecurity and Consumer Trust | Review 2 | 4
- Layer | Choice | Reason
- Language / data | Python, pandas, NumPy, SciPy, PyArrow | Standard data-science stack; Parquet storage
- Machine learning | scikit-learn, XGBoost | LR, RF, boosting, Isolation Forest, LOF, tuning
- Visualisation | Matplotlib, Seaborn, Plotly | Report figures, interactive notebooks
- Database | PostgreSQL 16 (SQLAlchemy) | Structured results and metadata
- Monitoring | Prometheus, Grafana, Loki + Promtail, cAdvisor | Metrics, dashboards, logs, container health
- Packaging | Docker Compose, Makefile, pytest | One-command, repeatable environment

Slide 6: Dataset collection

- FinTech Cybersecurity and Consumer Trust | Review 2 | 5
- Dataset | Source | Notes
- Transactions | Seeded synthetic generator | ~2% fraud, four stylised patterns expressed 75% of the time; injected defects
- Security events | Seeded synthetic generator | 18% attacks, six campaign bursts, benign background
- Survey | Synthetic placeholder (n~400) | 12 Likert items; replace with real consented responses
- Public benchmark | ULB credit-card fraud (optional) | Downloaded by user under its licence; not redistributed
- Synthetic data demonstrates the method; it is not evidence about real-world fraud prevalence.

Slide 7: Data preprocessing

- FinTech Cybersecurity and Consumer Trust | Review 2 | 6
- Step | Implementation | Why
- Validation | 13 checks: schema, ranges, domains, duplicates, missingness | Detect defects before modelling; exported as metrics
- Cleaning | Drop duplicate ids and invalid/missing amounts; row accounting | Every removal is counted and logged
- Features | Time, amount ratios vs customer's prior mean, recency, device/geo/IP flags, one-hot | History uses only earlier rows
- Survey privacy | Salted-hash IDs, banded demographics, k-anonymity (k=5) | Privacy-conscious handling
- Storage | Parquet + CSV in data/interim, data/processed | Typed, compact, portable

Slide 8: Initial implementation progress

- FinTech Cybersecurity and Consumer Trust | Review 2 | 7
- - src/ modules: data_collection, preprocessing, feature_engineering, database, metrics
- - Unit tests cover validation, cleaning, leakage-freedom (changing a future row leaves earlier features unchanged) and reproducibility
- - database/schema.sql with results, metadata, quality-check and trust tables
- - docker-compose.yml with health-checked service ordering
- - Pending: model training, anomaly detection, trust scoring, dashboards (Review 3)

- Evidence: repository code and tests. No model performance is reported at this stage.

Slide 9: Evaluation plan (fixed before experiments)

- FinTech Cybersecurity and Consumer Trust | Review 2 | 8
- Aspect | Plan
- Metrics | Accuracy (reported, not trusted), precision, recall, F1, ROC-AUC, PR-AUC, confusion matrix, false-positive rate
- Uncertainty | Bootstrap 95% CI; McNemar test for paired model comparison
- Trust | Cronbach's alpha, Spearman, OLS, Kruskal-Wallis, Mann-Whitney; Bonferroni across hypotheses
- Reproducibility | Seeds, data hash and library versions stored with each model

Review 3 - Machine learning, anomaly detection, trust scoring, dashboards

Slide 1: FinTech Cybersecurity and Consumer Trust

- Monthly Project Review 3 - Machine Learning, Anomaly Detection, Trust Scoring and Dashboards
- Final-year research & development project
- [Student name] - [Register no.] - Guide: [Guide name] - [Department / Institution]
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Slide 2: Progress since Review 2

- FinTech Cybersecurity and Consumer Trust | Review 3 | 1
- - Supervised models implemented: Logistic Regression, Random Forest, XGBoost with randomised hyper-parameter search
- - Unsupervised detectors implemented: Isolation Forest, Local Outlier Factor
- - Risk tiers, consumer trust scoring and statistics helpers implemented
- - Prometheus exporter, PostgreSQL persistence and 7 provisioned Grafana dashboards developed
- - Preliminary experiments run; results are first-run numbers on synthetic data, not final

Slide 3: Machine learning implementation

- FinTech Cybersecurity and Consumer Trust | Review 3 | 2
- Model | Role | Imbalance handling | Tuned parameters
- Logistic Regression | Interpretable baseline | class_weight=balanced | C
- Random Forest | Non-linear bagging | balanced_subsample | max_depth, min_samples_leaf, max_features
- XGBoost | Boosted trees | scale_pos_weight | depth, learning rate, subsample, colsample, min_child_weight
- Isolation Forest / LOF | Label-free detectors | contamination parameter | none (fit without labels)
- Protocol: randomised search (average precision) on train; F1-optimal threshold on validation; test used once.

Slide 4: Fraud and anomaly detection flow

- FinTech Cybersecurity and Consumer Trust | Review 3 | 3
- Cleaned transactions
- Features (40)
- Train/validation/test (chronological)
- Supervised models
- Unsupervised detectors
- Metrics + risk tiers
- Security events -> Isolation Forest
- Prometheus + PostgreSQL

Slide 5: Preliminary experiment: PR-AUC and ROC-AUC (test set)

- FinTech Cybersecurity and Consumer Trust | Review 3 | 4
- Real first-run output (run run_20261109_061947_57b9). Tuning budget n_iter is small; test set has 174 fraud cases, so estimates are noisy.

Slide 6: Preliminary experiment: threshold metrics

- FinTech Cybersecurity and Consumer Trust | Review 3 | 5

- Model | Precision | Recall | F1 | FPR
- logistic_regression | 0.883 | 0.563 | 0.688 | 0.0015
- random_forest | 0.928 | 0.667 | 0.776 | 0.0010
- xgboost | 0.982 | 0.626 | 0.765 | 0.0002
- isolation_forest | 0.365 | 0.437 | 0.398 | 0.0150
- local_outlier_factor | 0.187 | 0.287 | 0.226 | 0.0248
- Synthetic data: use these to validate the pipeline, not to claim real-world performance.

Slide 7: Consumer trust scoring (synthetic survey)

- FinTech Cybersecurity and Consumer Trust | Review 3 | 6
- Quantity | Value
- Respondents | 355
- Mean Consumer Trust Score (0-100) | 56.6
- Platform Security Index (0-100) | 79.8
- Consumer Trust Index (0-100) | 65.8
- Cronbach alpha (all items) | 0.906
- Min k-anonymity group | 5
- Survey data is synthetic placeholder data; the index is a proposed composite pending sensitivity analysis and real responses.

Slide 8: Experimental work: ROC and precision-recall curves

- FinTech Cybersecurity and Consumer Trust | Review 3 | 7
- Generated by src/model_evaluation/evaluate.py from the held-out test partition.

Slide 9: Grafana dashboard development

- FinTech Cybersecurity and Consumer Trust | Review 3 | 8
- Dashboard | Data sources | Status
- Research Overview | Prometheus, Loki | Provisioned
- Transaction & Fraud Analytics | Prometheus, PostgreSQL | Provisioned
- Cybersecurity Threat Monitoring | Prometheus, PostgreSQL, Loki | Provisioned
- Consumer Trust Indicators | Prometheus, PostgreSQL | Provisioned
- ML Model Performance | Prometheus, PostgreSQL | Provisioned
- Data Pipeline Monitoring | Prometheus, PostgreSQL, Loki | Provisioned
- Docker Infrastructure Health | Prometheus (cAdvisor) | Provisioned
- Panels query metrics computed by the pipeline and a held-out replay through the trained model; no hardcoded values.

Slide 10: Issues, risks and next steps

- FinTech Cybersecurity and Consumer Trust | Review 3 | 9
- - Increase tuning budget and repeat on several seeds for stable estimates
- - Run the optional public credit-card benchmark
- - Replace synthetic survey with real, ethics-approved responses
- - Finish sensitivity analysis of the trust index; write results chapters (Review 4)

Review 4 - Results, comparison, dashboards, conclusions

Slide 1: FinTech Cybersecurity and Consumer Trust

- Monthly Project Review 4 - Results, Model Comparison, Dashboards and Conclusions
- Final-year research & development project
- [Student name] - [Register no.] - Guide: [Guide name] - [Department / Institution]
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Slide 2: Completed implementation

- FinTech Cybersecurity and Consumer Trust | Review 4 | 1
- Component | Status
- Pipeline (collect to report, 14 timed stages) | Complete
- Models: LR, RF, XGBoost, Isolation Forest, LOF | Complete
- Risk tiers, trust scoring, statistics | Complete
- PostgreSQL schema, views, persistence | Complete
- Prometheus exporter, Loki logging, 7 Grafana dashboards, alert rules | Complete
- 16 notebooks, pytest suite, Docker Compose, Makefile | Complete
- Real survey data / ethics approval | Pending - bundled survey is synthetic

Slide 3: Final system architecture

- FinTech Cybersecurity and Consumer Trust | Review 4 | 2
- Docker network: fintech-net (ports bound to localhost)
- Worker
- Python pipeline
- + /metrics :8000
- JupyterLab
- :8888
- PostgreSQL
- results + metadata
- Promtail ->
- Loki (logs)
- Prometheus
- :9090
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Slide 4: Model comparison: PR-AUC, ROC-AUC, F1 (test set)

- FinTech Cybersecurity and Consumer Trust | Review 4 | 3
- Run run_20261109_061947_57b9, seed 42. Champion (highest PR-AUC among supervised): xgboost.

Slide 5: Performance metrics (test set)

- FinTech Cybersecurity and Consumer Trust | Review 4 | 4

- Model | Accuracy | Precision | Recall | F1 | ROC-AUC | PR-AUC | FPR
- logistic_regression | 0.9901 | 0.883 | 0.563 | 0.688 | 0.872 | 0.666 | 0.0015
- random_forest | 0.9925 | 0.928 | 0.667 | 0.776 | 0.903 | 0.741 | 0.0010
- xgboost | 0.9925 | 0.982 | 0.626 | 0.765 | 0.885 | 0.743 | 0.0002
- isolation_forest | 0.9743 | 0.365 | 0.437 | 0.398 | 0.780 | 0.400 | 0.0150
- local_outlier_factor | 0.9618 | 0.187 | 0.287 | 0.226 | 0.683 | 0.116 | 0.0248
- Accuracy is high for any model on ~2% fraud and is not a selection criterion; PR-AUC and FPR are.

Slide 6: Statistical comparison

- FinTech Cybersecurity and Consumer Trust | Review 4 | 5
- Item | Result
- Champion | xgboost
- Champion ROC-AUC 95% bootstrap CI | [0.847, 0.918]
- McNemar champion vs Logistic Regression | chi2=11.61, p=0.0007
- Confusion (champion) | TP=109 FP=2 FN=65 TN=8777

Slide 7: Risk tiers (champion, test set)

- FinTech Cybersecurity and Consumer Trust | Review 4 | 6
- Tier | Transactions | Frauds | Observed fraud rate | Action
- low | 3 | 0 | 0.000 | allow
- medium | 7871 | 37 | 0.005 | step-up authentication
- high | 912 | 12 | 0.013 | hold for manual review
- critical | 167 | 125 | 0.749 | block + alert

Slide 8: Confusion Matrices

- FinTech Cybersecurity and Consumer Trust | Review 4 | 7
- Confusion matrices at validation-tuned thresholds.

Slide 9: Consumer Trust Overview

- FinTech Cybersecurity and Consumer Trust | Review 4 | 8
- Consumer trust (synthetic survey): score distribution, dimensions, correlations.

Slide 10: Security Overview

- FinTech Cybersecurity and Consumer Trust | Review 4 | 9
- Security events and attack types.

Slide 11: Security and trust results

- FinTech Cybersecurity and Consumer Trust | Review 4 | 10
- Quantity | Value
- Attack block rate | 0.797
- Isolation Forest ROC-AUC vs known attacks | 0.994
- Mean Consumer Trust Score | 56.6
- Platform Security Index | 79.8
- Consumer Trust Index | 65.8
- Cronbach alpha | 0.906
- Survey is synthetic; trust values demonstrate the method and are not findings about real consumers.

Slide 12: Grafana dashboard demonstration

- FinTech Cybersecurity and Consumer Trust | Review 4 | 11
- - Live demo: Overview, Pipeline Monitoring, Fraud, Model Performance, Trust, Infrastructure
- - Add screenshots as review/assets/grafana_*.png and re-run review/scripts/build_all.py
- No screenshots were available when this deck was generated.

Slide 13: Conclusions

- FinTech Cybersecurity and Consumer Trust | Review 4 | 12
- - A reproducible, observable platform links fraud/anomaly detection with a consumer trust measure (3 supervised + 2 unsupervised detectors)
- - Tree ensembles are compared with a linear baseline using PR-AUC, recall and false-positive rate rather than accuracy
- - Unsupervised detectors are useful without labels but are compared separately from supervised models
- - Grafana shows live computed metrics, logs and database results for the entire experiment

Slide 14: Limitations

- FinTech Cybersecurity and Consumer Trust | Review 4 | 13
- Data and validity
- - Synthetic transactions encode our assumptions
- - Bundled survey is synthetic
- - Few positives in test set widen uncertainty
- - No concept drift modelled
- Method
- - Trust Index is a proposal, not validated
- - Single dataset family and limited tuning budget
- - Replay stream is not production traffic
- - Alertmanager not deployed

Slide 15: Future enhancements

- FinTech Cybersecurity and Consumer Trust | Review 4 | 14
- - Run on the public ULB dataset and other licensed data; repeat over many seeds
- - Real ethics-approved survey; factor analysis and index validation
- - Add calibration, cost-sensitive thresholds, explainability (SHAP)
- - Drift monitoring and scheduled retraining; Alertmanager notifications
- - Combine unsupervised scores with supervised features

What changed between reviews

From -> to	Added
1 -> 2	Concrete architecture, stack, datasets, preprocessing, implemented foundations and tests
2 -> 3	Model training, anomaly detection, trust scoring, dashboards and preliminary results
3 -> 4	Final comparison, statistics, risk tiers, security/trust results, conclusions, limitations, future work